YUANBO PANG

UC Berkeley EECS | Foundation Agents & Robotics
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Education

University of California, Berkeley *Expected Dec 2026*
B.S. Electrical Engineering & Computer Sciences GPA: 3.762/4.0 (in-major: 3.811)
Relevant coursework: CS 170, CS 168, CS 184, EECS 106A, CS 61B, CS 70

Stanford University (Summer Session) — CS 229: Machine Learning *Summer 2025*

De Anza College — Computer Science, GPA: 4.0/4.0 *2023 – 2025*

Research Experience

Research Assistant, Berkeley RDI / BAIR (Prof. Dawn Song) *Oct 2025 – Present*
Agents' Last Exam — co-first author on V1; leading two research tracks in V2

- **Foundation Agent / 3D spatial understanding.** Building training and evaluation environments in simulation (Blender, NVIDIA Isaac Lab) where agents actively move the viewpoint to explore a scene and judge physical state — 3D understanding through exploration, not inference from a fixed 2D view.
- **Ultra-long-horizon engineering.** Constructing tasks from large-scale production software that human teams measure in months to years, with a harness that evaluates *teams* of agents — coordination, division of labor, and context handoff at horizons far beyond single-agent benchmarks.
- For V1, built task/environment pipelines, computer-use and tool interfaces, scalable GCP execution, and programmatic evaluators producing verifiable rewards for RLVR; now run ALE evaluations with frontier model teams (OpenAI, Anthropic, Meta TBD Lab, Qwen, Seed), analyzing agent trajectories and reporting failure modes back to the labs.

Robotics Research, Prof. Masayoshi Tomizuka's Lab, UC Berkeley *2026 – Present*
Large-Scale Robot-Learning Data Benchmark — mentored by Zhuo Xu

- Built `roboqa`: dataset-quality benchmark at EgoDex scale (338K episodes, 829 h) — health, diversity, and trainability scored separately; every score published with full provenance.
- Agent workflows inside processing and annotation: VLM-judged quality audits; raw captures auto-converted to a canonical, embodiment-tagged format (egocentric hand, gripper, humanoid).
- Train and post-train VLAs on 500 h of curated data; score the resulting policies in simulation.

URAP Research Assistant, Prof. Avideh Zakhor, UC Berkeley EECS *Spring 2026 – Present*

- Building a high-quality UAV-to-UAV dataset and PyTorch pipelines for zero-shot detection and multi-object tracking under high-speed ego-motion; integrating optical flow with YOLO and SAM2-style baselines. Targeting CVPR workshop submission.

Engineering Experience

Equile — independent side project: "the dry lab for AI scientists" *Apr 2026 – Oct 2026*

- Sole owner of the runtime and control plane: serverless hosting for agent harnesses (Claude Code, Codex, Cursor, Grok) — isolated per-session sandboxes, multi-tenant packing that cuts per-agent compute >20×, snapshot pause/resume, public SDK/CLI.

Agentic Tech Staff, Starbot *Nov 2025 – May 2026*

- Led the agentic stack for autonomous service robots: vision-language-action models, agentic planning and control, multi-robot coordination.
- Shipped as the CES 2026 demo system, deployed on the show floor 8 hours/day across coffee preparation, popcorn making, and front-desk reception; covered live by Japan's NHK News, demoed to 1,500+ attendees, and converted into 50+ inbound customer conversations.

Software Engineering Intern, Geopogo *Fall 2025*

- Shipped a production multimodal-generation workflow (React/TypeScript, FastAPI, Gemini, Firebase, CI/CD).

Research Interests

Foundation Agents: 3D and environment understanding; RL systems for agent–environment interaction

Robotics: 3D understanding to strengthen world-action models (VLA)

Publications

Agents' Last Exam.
Yiyou Sun*, Xinyang Han*, Weichen Zhang*, **Yuanbo Pang***, et al., Dawn Song. *Under review at NeurIPS 2026.* arXiv:2606.05405. (*Co-first author & core contributor; 310 authors total)
#1 Paper of the Week, Hugging Face Daily Papers (Jun 2026).

ALE-Replica: Last Exam for Agents in Reconstructing Professional Software.
Yuanbo Pang (first author), et al. *Under double-blind review at CVPR 2027.* Agents rebuild professional applications (AutoCAD, Maya, Blender, Ansys) from documentation; evaluation checks state and GUI after every operation against native execution.

ScienceIDE: Turning World's Scientific Codebase into Agent Learnable Environments.

Hejia Geng*, Zesen Huang*, Haoyang Li, Wenbin Li, Koutian Wu, Zihan Zhou, **Yuanbo Pang**, et al., Ling Yang. arXiv:2609.19134. (AItonomy Foundation; sponsors: PhAI-Labs, Qwen)
#2 Paper of the Day, Hugging Face Daily Papers (Sep 2026).

Agent's GameDev Exam.
Zonglin Di, Yinhe Wang, **Yuanbo Pang**, Yiyou Sun, Weichen Zhang, Xinyang Han, et al., Dawn Song. *Under double-blind review at ICLR 2027.*

Awards

Inspire Fellowship, rednote (Xiaohongshu) — inaugural 2026 cohort, one of the first 6 fellows selected worldwide; $100K grant.
HackMIT Modal Sponsor Prize (2025); Smartest AI Agent, HackMIT (2024).

Technical Skills

ML / RL: PyTorch, reinforcement learning (RLVR), VLA, model finetuning & evaluation
Vision / Robotics / Sim: YOLO, SAM2, optical flow, Isaac Lab, Blender
Agent / Systems: benchmarks, automated evaluators, tool-use & MCP, Docker, GCP, Modal, FastAPI, Linux